

# Management of Fear & Stress



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# Outline



- Lecture Learning Outcomes
- Introduction
- Definitions: Stress, dental care related fear and anxiety
- *Etiology of Dental Care-Related Fear and Anxiety*
- *Classification of Patients with Dental Care-Related Fear and Anxiety*
- Assessment of Dental Care-Related Fear and Anxiety
- Interventions for Dental Care-Related Anxiety, Fear, and Phobia

# Lecture learning outcomes



- Outline concepts of dental fear, stress and anxiety associated with oral health care
- Describe the techniques for reducing dental fear

# Introduction - STRESS



- The term “stress” should be restricted to conditions where an environmental demand exceeds the natural regulatory capacity of an organism, in particular, situations that include unpredictability and uncontrollability.
- The interactional model of stress, suggesting that a mismatch between demand and ability leads to stress which has physiological, behavioural and cognitive effects.

# DENTAL CARE-RELATED FEAR



- A distressing **emotional response** to treatment related stimuli, typically characterized by **physiological responsivity**, feelings and reports of apprehension and/or dismay and avoidance. A severe fear response may involve panic symptoms or even a panic attack in the dental clinic.
- Components that predispose a patient to panic attack are (a) an inability to escape easily; (b) a reduction in controllability; and (c) a reduction in predictability.



# DENTAL CARE-RELATED ANXIETY



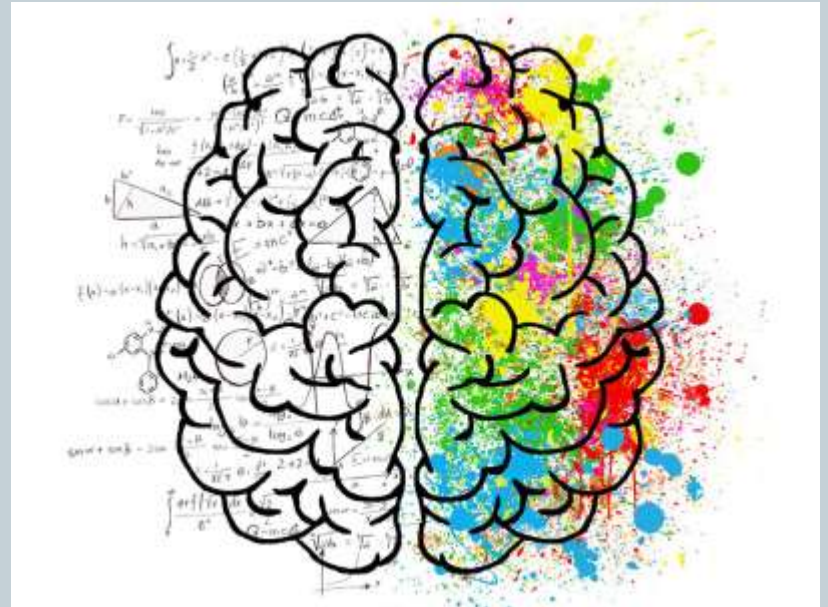
- A more **cognitively involved emotional response** to stimuli or experiences associated with dental treatment.
- Dental patients are most likely to experience anxiety well prior to their dental visit as they consider elements of the dental visit that have caused or could cause discomfort, and this anxiety may wax and wane between appointments.

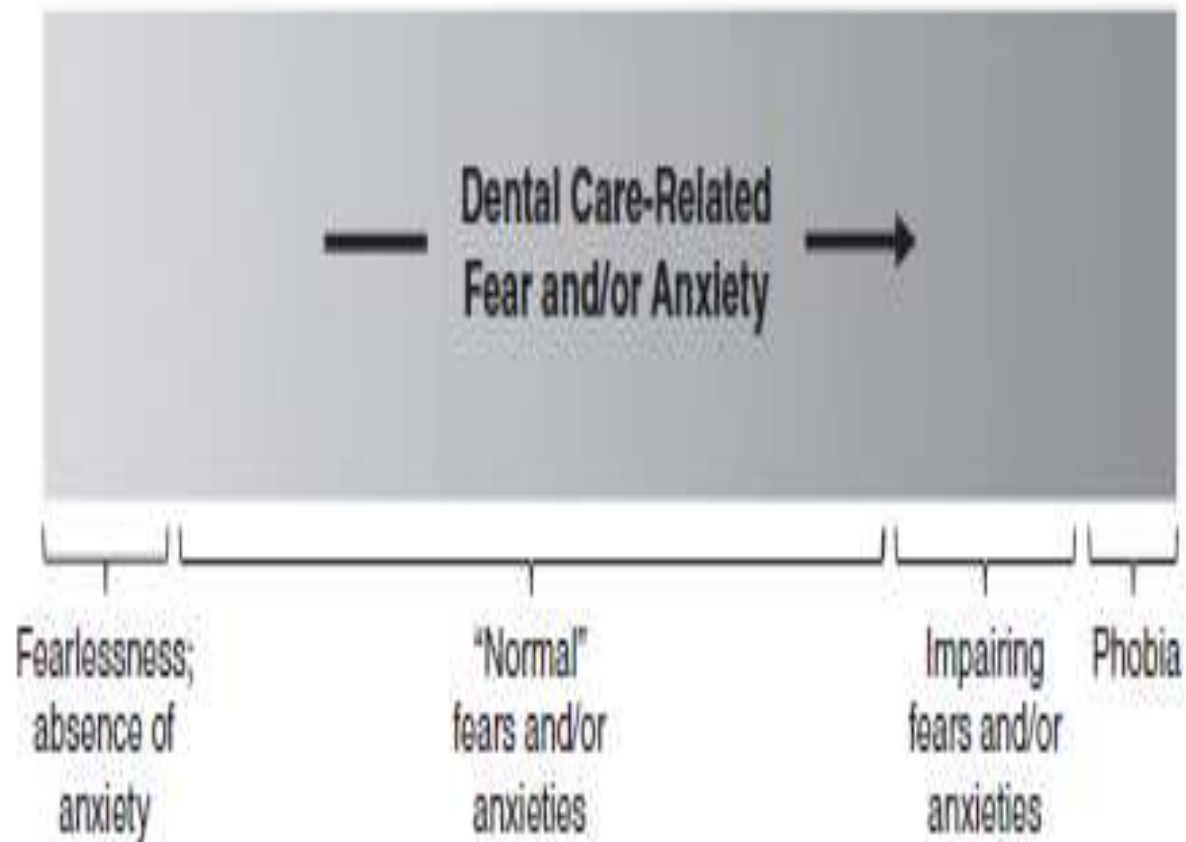


# Components of Anxiety



1. Somatic (experienced predominantly in a bodily fashion)
2. Cognitive (experienced predominantly in a mental fashion).





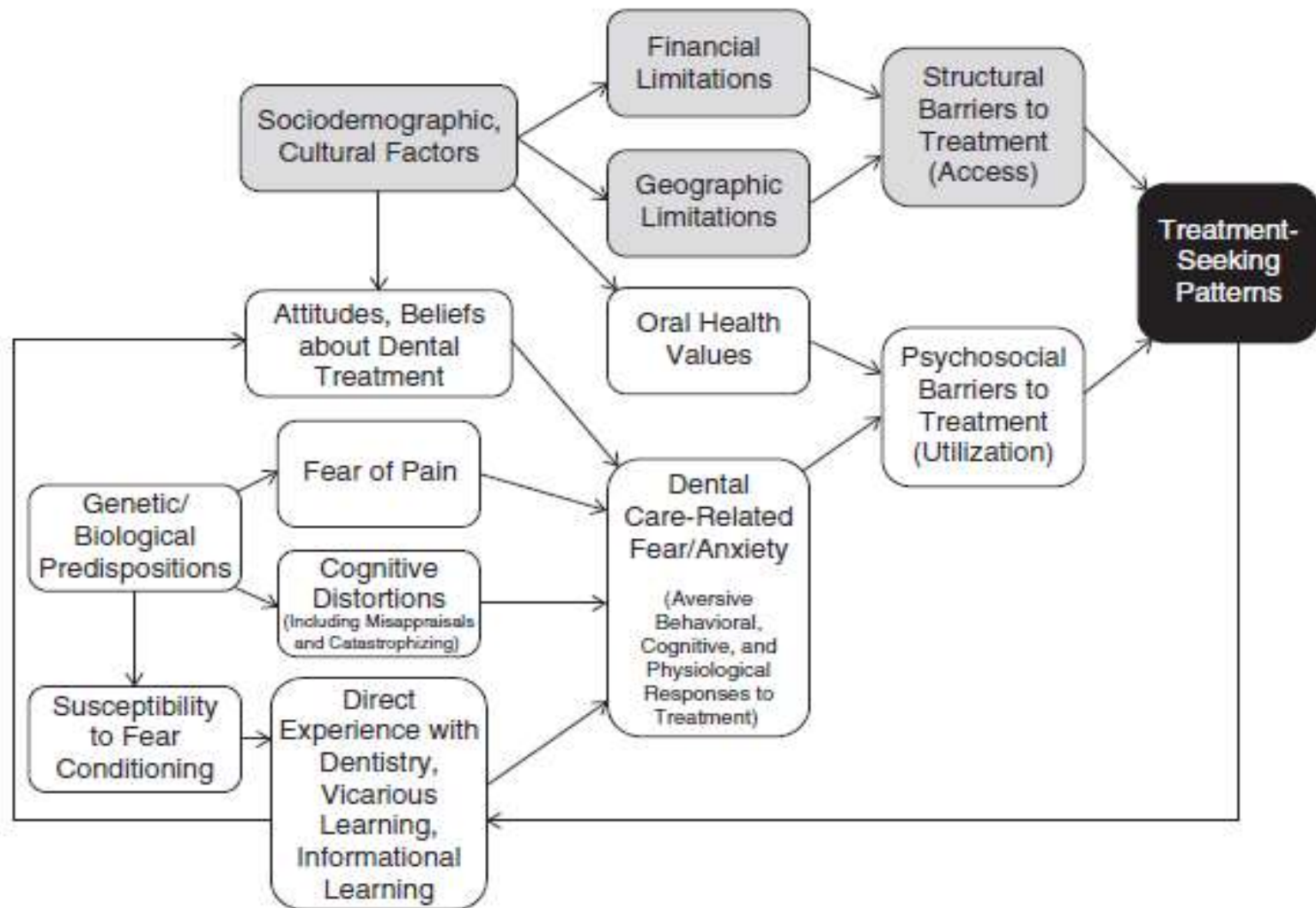
**Fig. 12.1.** Model of the continua of dental care-related anxiety and dental care-related fear across the general population.



# *Etiology of Dental Care-Related Fear and Anxiety*



1. **Conditioning** (experiencing a traumatic event will lead us to believe similar future events may also be traumatic)
2. **Social/observational Learning** (dental treatment is uncomfortable, distressing, or painful via witnessing emotional arousal in others, information sharing)
3. **Biological Influences** (biological and genetic factors)
4. **Cognitive Factors** (misperceptions, negative thoughts, and patterns of thinking that include worries about danger or lack of control)
5. **Personality** (personality features, such as being easily stressed, experiencing strong negative emotions, and being less sociable)
6. **Pain-Related Factors** (Differential pain sensitivity, perceptions of or fears about pain)



**Fig. 12.2.** Conceptual model of the impact of dental care-related fear and anxiety on treatment-seeking patterns, and relation to associated variables.

# *Classification of Patients with Dental Care-Related Fear and Anxiety*



- Two etiological categories: exogenous or endogenous.
  1. Exogenous dental care-related anxiety are thought to have developed anxiety/fear as a result of conditioning through traumatic experiences or observational learning.
  2. Endogenous dental care-related anxiety/fear are thought to have developed dental-specific distress as a result of some other underlying mood or anxiety disorder.

# Assessment of Dental Care-Related Fear and Anxiety

1. Verbal report
2. Behavioral
3. Physiological



# 1. Verbal Report



- Self-report instruments; Questionnaire: example: The Dental Fear Survey (DFS; is a 20-item self report measure of fearful and anxious reactions).

| Dental Fear Survey Questions                                                          |
|---------------------------------------------------------------------------------------|
| 1 Has fear of dental work ever caused you to put off making an appointment?           |
| 2 Has fear of dental work ever caused you to cancel or not appear for an appointment? |
| 3 When having dental work done: My muscles become tense ...                           |
| 4 When having dental work done: My breathing rate increases ...                       |
| 5 When having dental work done: I perspire ...                                        |
| 6 When having dental work done: I feel nauseated and sick to my stomach ...           |
| 7 When having dental work done: My heart beats faster ...                             |
| 8 Making an appointment for dentistry ...                                             |
| 9 Approaching the dentist's office ...                                                |
| 10 Sitting in the waiting room ...                                                    |
| 11 Being seated in the dental chair ...                                               |
| 12 The smell of the dentist's office ...                                              |
| 13 Seeing the dentist walk in ...                                                     |
| 14 Seeing the anesthetic needle ...                                                   |
| 15 Feeling the needle injected ...                                                    |
| 16 Seeing the drill ...                                                               |
| 17 Hearing the drill ...                                                              |
| 18 Feeling the vibrations of the drill ...                                            |
| 19 Having your teeth cleaned ...                                                      |
| 20 All things considered, how fearful are you of having dental work done?             |

## 2. Behavioral

- Assess patients' overt behaviors: “nervous” behaviors, such as leg-swinging by highly anxious/fearful patients awaiting dental treatment, while in the waiting room.
- Facial expressions
- Amount and content of spontaneous speech (e.g., humor, question-asking), eye closures, use of the hands, speed at which a patient sits in the dental chair, asking for breaks during procedures, and many other behaviors may provide a bounty of data about dental care-related fear and anxiety.





- Most important overt behaviors: Avoidance, delaying appointments, or “escaping” from appointments by late cancellations or rescheduling.

### 3. *Physiological*



- Psychophysiological responses: Perspiration on the palms, forehead, or upper lip, increased respiration rate, high heart rate or blood pressure, skin conductance, forehead muscle tension; vary with the severity of dental fear/anxiety.



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# *Other Approaches to the Assessment of Dental Care-Related Fear and Anxiety*



- The Dental Fear Interview



# Interventions for Dental Care-Related Anxiety, Fear, and Phobia

- A key ingredient of each of these fear- or anxiety-reducing strategies is high-quality rapport.
- Demonstrating that the patient is heard and understood will convey empathy and compassion that can facilitate rapport-building, reduce acute dental care-related fear and/or anxiety, and prepare the patient for other interventions to prevent or manage dental care related fear and anxiety.

# Types of Interventions



- *I. Pharmacological Interventions*
- *II. Behavioral, Cognitive-Behavioral, and Educational Interventions*

# *I. Pharmacological Interventions*



- Nitrous oxide, anti-anxiety medications, sedation, or even general anesthesia may be necessary for the extremely fearful, anxious, or phobic patient.



## *II. Behavioral, Cognitive-Behavioral, and Educational Interventions*



- There is a host of interventions, such as psychoeducation, stimulus control, social praise, modeling, behavioral control, exposure therapy, relaxation, paced breathing, systematic desensitization, stress inoculation training, hypnosis, and biofeedback, which can be used to prevent or reduce dental care-related fear and anxiety.

# a. Behavioral Interventions



- “Stimulus control”
- Social praise
- Modeling
- Increasing control and predictability (‘tell-show-do’ technique).
- Systematic desensitization
- Paced breathing
- Utilizing **exposure therapy**: The patient has multiple visits to the clinic before actual treatment may begin.



# Systematic Desensitization



- A process in which a patient first learns a relaxation strategy, and then is gradually “exposed” to fear-evoking stimuli (e.g., the dental chair, sound of the drill), typically in a stepwise fashion.



## b. Cognitive Interventions



- Cognitive restructuring: Aims to facilitate a new understanding that the feared stimuli are unlikely to be dangerous and in turn that avoidance or other safety behaviours are not required.





## c. Psychoeducation



- For those individuals whose fear or anxiety seems related to uncertainty, not knowing what might happen during dental procedures, it may be especially helpful to interact with them by providing informational control that is, by **sharing information** in a psychologically sensitive fashion.
- A psychoeducational approach and can “set the stage” for a highly fearful/anxious patient to receive needed dental care is motivational interviewing.

# Chairside Techniques for Reducing Dental Fear



- According to the role of relaxation behavioral interventions can be classified as: (a) non-relaxation based techniques, (b) “quasi-relaxation”-based techniques, and (c) relaxation-based techniques.

# Non-relaxation-based techniques



- Intervention strategies that do not utilize relaxation types of procedures. Examples: communication skills (including the redefining of sensations), distraction, behavior modification, modeling, cognitive restructuring or cognitive behavior modification, stress inoculation, iatrosedation, and acupressure.



# Distraction



- Distraction is an easily utilized method of reducing a patient's apprehension during the occurrence of uncomfortable sensations.
- The purpose of distraction techniques is to refocus the patient's attention away from the potentially painful stimulus or procedure.
- Distraction techniques can be grouped into two main categories: physical methods and mental methods. Both are effective and can be employed separately or in combination.





- 1. Physical methods: Cheek-jiggling, leg-raising, listening to music or recorded books through earphones, playing video games, watching movies, virtual reality on monitors, rubbing the cheek of the patient or massaging their gums during injections, or simply asking the patient to count to the holes in the ceiling, breath control.
- 2. Cognitive methods: Include various “mental exercises” that the patient can engage in while the practitioner is performing particularly stressful parts of a procedure.



# Quasi-relaxation Based Techniques



- Are those that achieve relaxation indirectly, without employing a formal progressive muscle relaxation procedure.
- Examples are guided imagery, paced breathing, biofeedback, graduated exposure, and meditation.



# Relaxation-based techniques



- Incorporate progressive relaxation-type procedures directly as part of the intervention.
- Examples: Progressive relaxation, systematic desensitization, and hypnosis.

Both systematic desensitization and hypnosis are often recommended primarily for extremely anxious or phobic patients, and are often accomplished in the office of a mental health practitioner to whom the patient has been referred.



- Techniques chosen to be used should reflect the degree of fear the patient is experiencing.
- **For mild anxiety:** Simple non-relaxation procedures such as distraction may be adequate.
- In general, complexity increases from non relaxation- based techniques to relaxation-based techniques.





- Pharmacological approaches are essential in helping anxious, fearful, and/or phobic dental patients receive needed care, and can be used in an integrated fashion with psychoeducational, behavioral, and cognitive interventions.

# References



- Chapters 3,10,12: Mostofsky DI, Fortune F. Behavioral dentistry. John Wiley & Sons; 2013 Dec 31.



Thank You